FABER-CASTELL
Date
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Lecture - 30
Regression, Regularizn & Model Selection

- LS → Statistical Prop.

$\varepsilon \sim N(0, \sigma^2 I)$

OLS ← $\hat{\beta} \sim N(\beta, (X^T X)^{-1} \sigma^2)$ → estimator distribu^n

Var. estimate $\hat{\sigma}^2 = \frac{1}{n-p-1} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$

$E[\hat{\beta}] = \beta$ (unbiased) ; $E[\hat{\sigma}^2] = \sigma^2$

$(n-p-1)\hat{\sigma}^2 \sim \sigma^2 \chi^2_{n-p-1}$

- Gradient descn^

$L(\beta) = \frac{1}{n}(y - X\beta)^T (y - X\beta)$

$= \frac{1}{n}(y^T y - 2\beta^T X^T y + \beta^T X^T X \beta)$

$\nabla L(\beta) = \frac{-2}{n} X^T y + \frac{2}{n} X^T X \beta$

$= \frac{-2}{n} X^T (y + X\beta)$

$\beta_{t+1} = \beta_t - \eta \nabla L(\beta_t)$

$$\boxed{\beta_{t+1} = \beta_t + \frac{2\eta}{n} X^T (y - X\beta_t)}$$

$r_t = y - X\beta_t$ → residual

- with p predictors, we have 2^p possible subsets
- goal

$$\min_{\beta} \| y - X_S \beta_S \|^2$$

$S \subset \{1, \ldots, p\}$ subset of var. $k = |S|$

$k \uparrow$, train error ↓

we have to find a way to get optimal k.

- for each $k = 0, 1 \dots p$

$$S_k = \underset{|S| = k}{\arg\min} \| y - X_S \beta_S \|^2$$

search all 2^p subsets, returns best model for each size k.
Training RSS ↓ with k.
large k → low bias, high var.
small k → high bias, low var.
computationally infeasible for large p

- stepwise methods
forward → intercept, add var. that most reduces RSS, produce nested seq.
backward → start with full set, remove least useful var, req. $n > p$
forward stagewise → adds very small coeff., extremely conservative, useful in high-dim.

- Model selec^n at 2^p ⇒ unstable
(low bias, high var.)
⇓
sol. → shrinkage

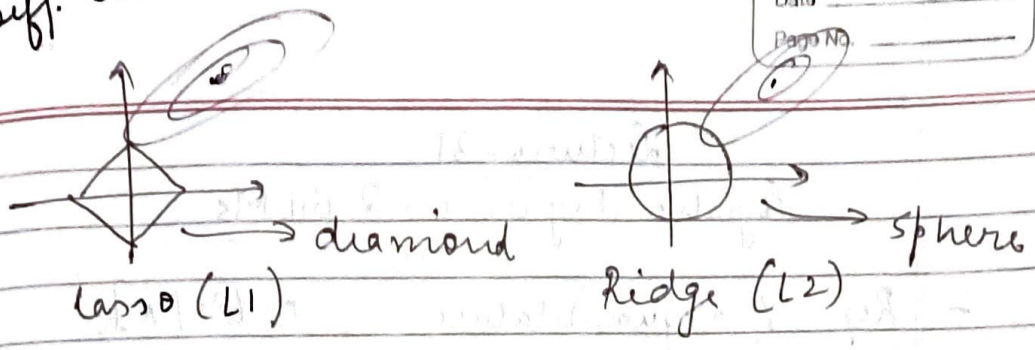
$$\min_{\beta} \| y - X\beta \|^2 + \lambda \Omega(\beta)$$

shrink coeff. towards 0, red. var. w/o hard selec^n, continuous models

- Ridge $\Omega(\beta) = \sum \beta_j^2$
→ shrink coeff. cont., no exact 0s.
→ shrinkage

- Lass $\Omega(\beta) = \sum |\beta|$
→ prod. sparse sol., perform var. selec^n
→ shrinkage + selec^n

coeff. exact 0.



- Ridge →

$$\hat{\beta}_R = \arg\min_{\beta} \|y - \beta X\|^2 + \lambda \Sigma \beta_j^2$$

dist wts. across correlated var. (low var.)

or $\min_{\beta} \|y - X\beta\|^2$ s.t. $\Sigma \beta_j^2 \leq t$

$$\boxed{\hat{\beta}_R = (X^T X + \lambda I)^{-1} X^T y}$$

adds λI → stabilize inversion, handles multicollinearity

- Lasso →

selects, discard others interpretable sparse model

$$\hat{\beta}_L = \arg\min_{\beta} \|y - X\beta\|^2 + \lambda \Sigma |\beta_j|$$

$\Sigma |\beta_j| < t$

non-differentiable at $\beta_j = 0$, leads to exact 0 sparse model ⇒ automatic var. selec^n